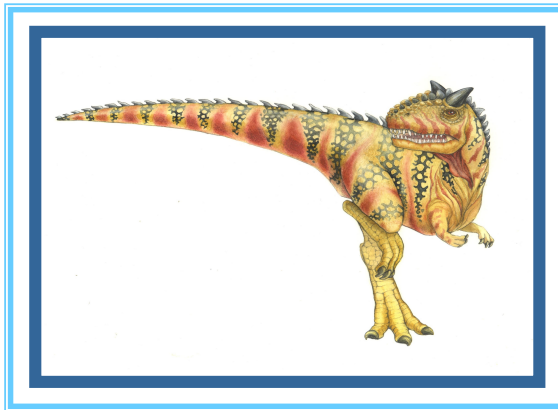


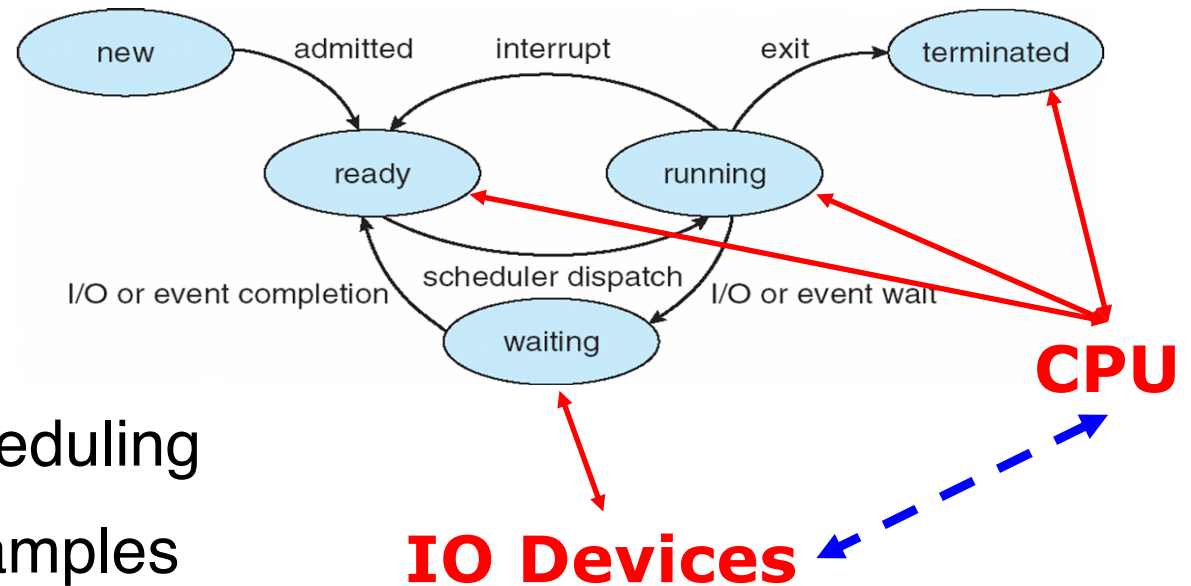
Chapter 5: CPU Scheduling





Chapter 5: CPU Scheduling

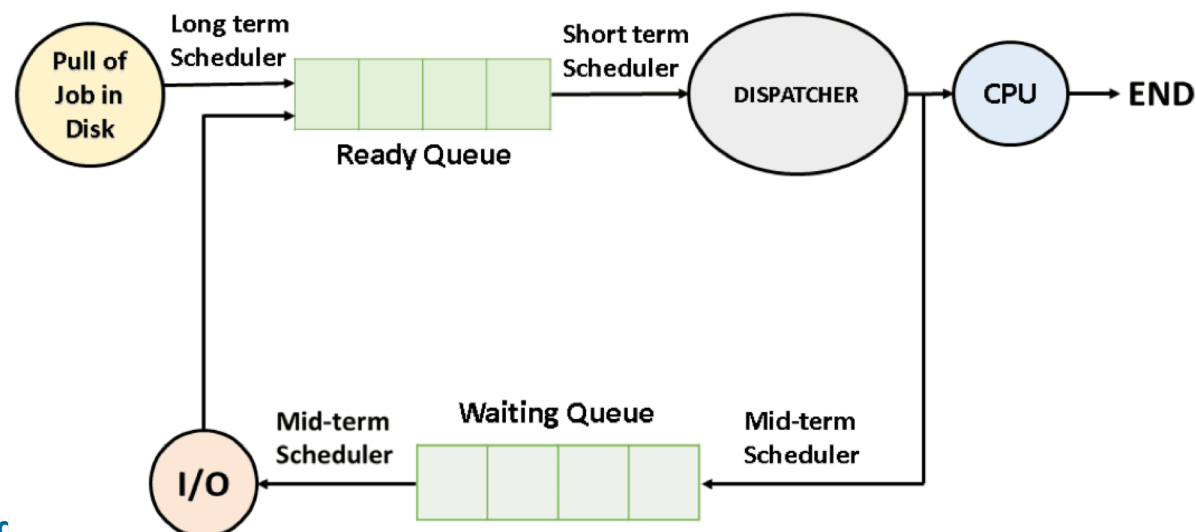
- Basic Concepts
- Scheduling Criteria
- Scheduling Algorithms
- Thread Scheduling
- Multiple-Processor Scheduling
- Operating Systems Examples
- Algorithm Evaluation





Objectives

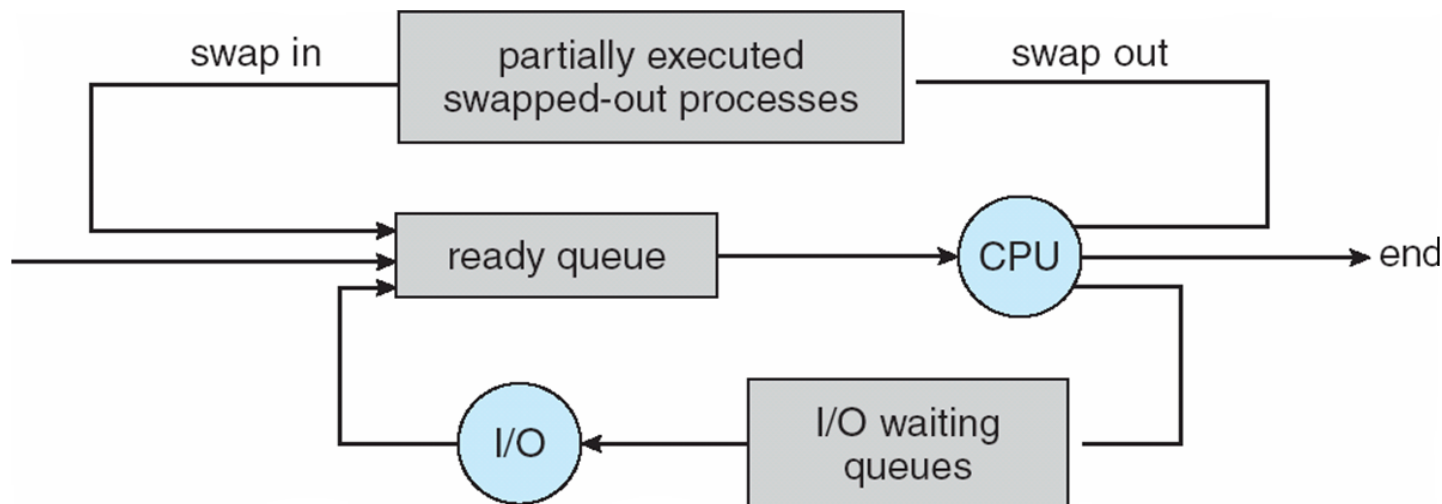
- To introduce CPU scheduling, which is the basis for multi-programmed operating systems
- To describe various CPU-scheduling algorithms
- To discuss evaluation criteria for selecting a CPU-scheduling algorithm for a particular system





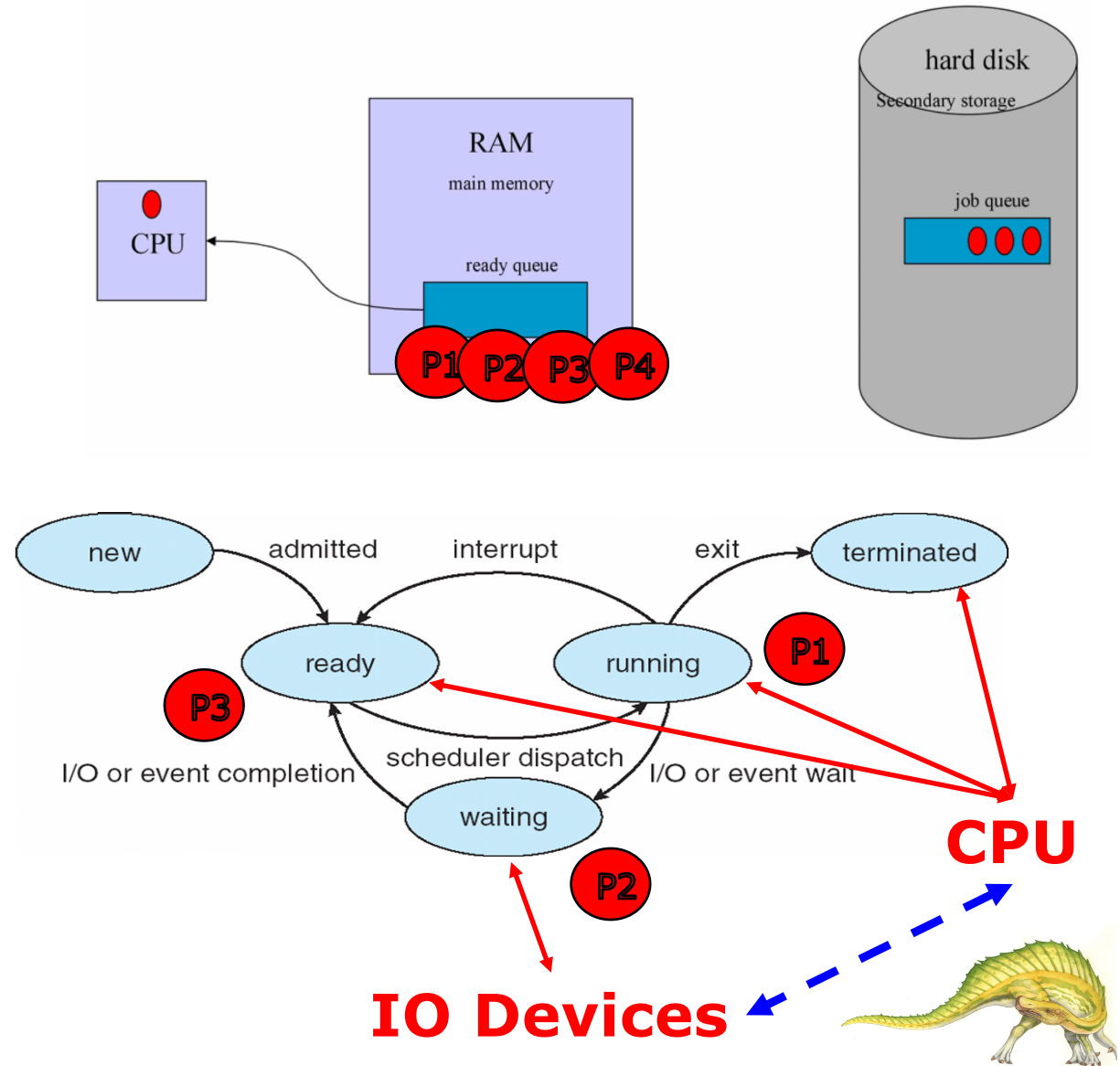
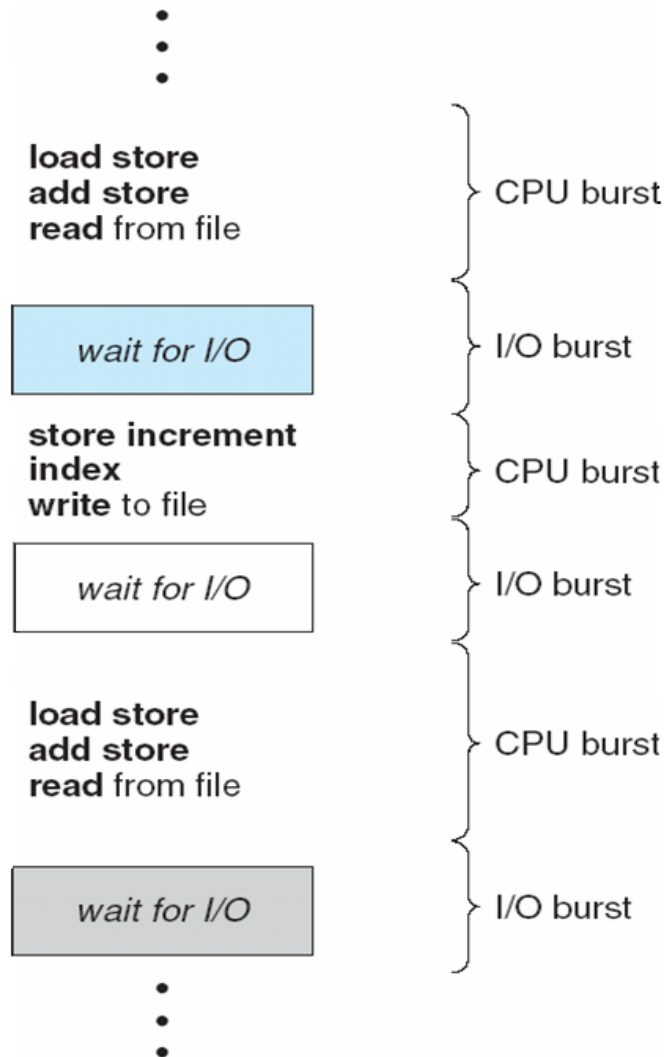
Basic Concepts

- Maximum CPU utilization obtained with multiprogramming
- CPU–I/O Burst Cycle – Process execution consists of a *cycle* of CPU execution and I/O wait
- **CPU burst** distribution



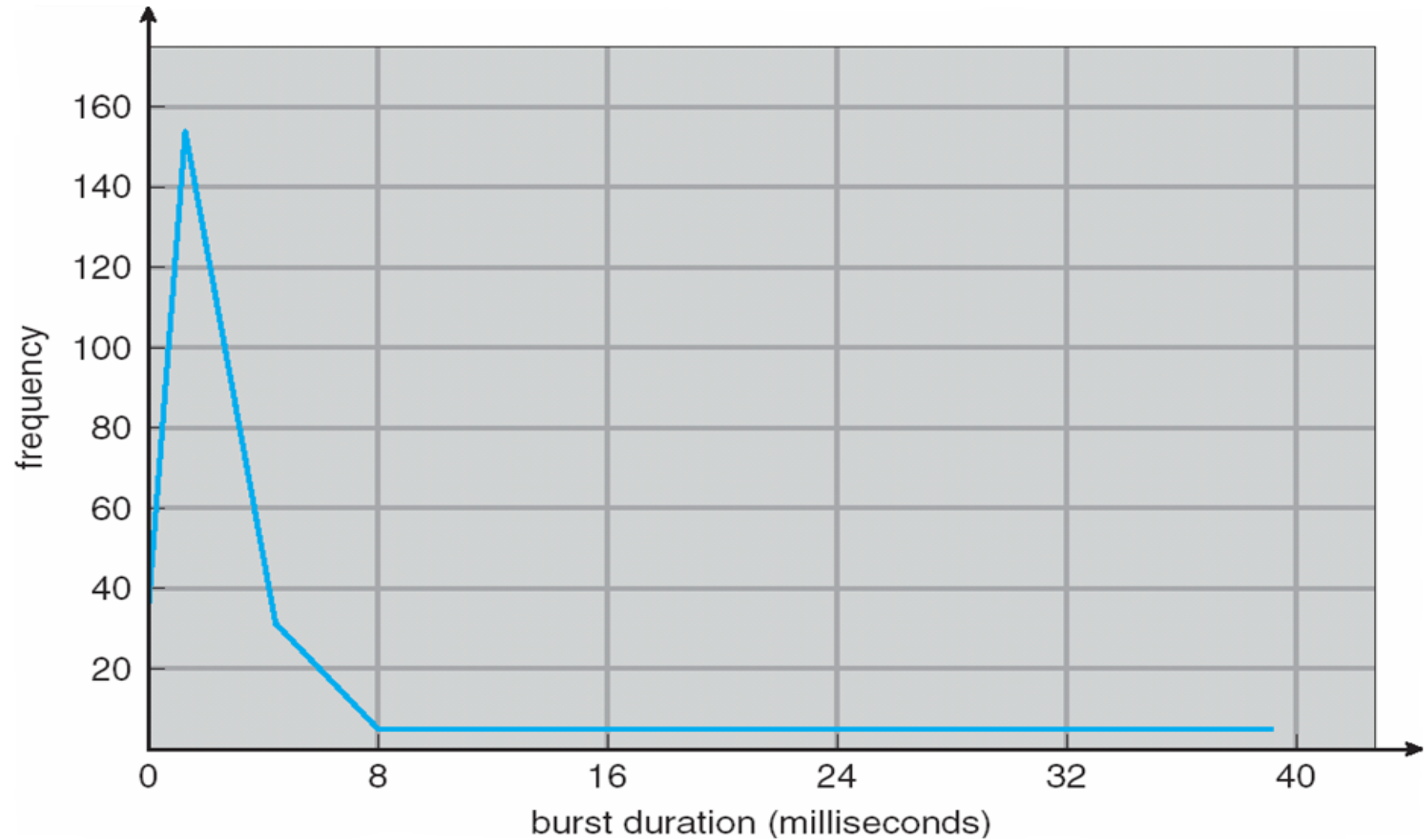


Alternating Sequence of CPU and I/O Bursts





Histogram of CPU-burst Times





CPU Scheduler

- Selects from among the processes in ready queue, and allocates the CPU to one of them
 - Queue may be ordered in various ways
- CPU scheduling decisions may take place when a process:
 1. Switches from running to waiting state
 2. Switches from running to ready state
 3. Switches from waiting to ready
 4. Terminates
- Scheduling under 1 and 4 is **nonpreemptive**
- All other scheduling is **preemptive**
 - Consider access to shared data
 - Consider preemption while in kernel mode
 - Consider interrupts occurring during crucial OS activities





Dispatcher

- Dispatcher module gives control of the CPU to the process selected by the short-term scheduler; this involves:
 - switching context
 - switching to user mode
 - jumping to the proper location in the user program to restart that program

- **Dispatch latency** – time it takes for the dispatcher to stop one process and start another running





Scheduling Criteria

- **CPU utilization** – keep the CPU as busy as possible
- **Throughput** – # of processes that complete their execution per time unit
- **Turnaround time** – amount of time to execute a particular process
- **Waiting time** – amount of time a process has been waiting in the ready queue
- **Response time** – amount of time it takes from when a request was submitted until the first response is produced, not output (for time-sharing environment)





Scheduling Algorithm Optimization Criteria

- Max CPU utilization
- Max throughput
- Min turnaround time
- Min waiting time
- Min response time





Scheduling Algorithms

- First-Come, First-Served (FCFS)
- Shortest Job First (SJF)
- Shortest Remaining Time First (SRTF)
- Round-Robin (RR)
- Priority Scheduling
- Highest Response Ratio Next (HRRN)
- Multilevel Queue
- Multilevel Feedback Queue

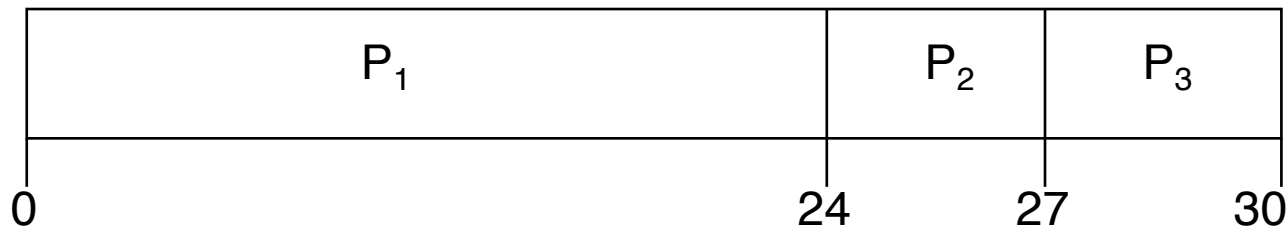




First-Come, First-Served (FCFS) Scheduling

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- Suppose that the processes arrive in the order: P_1 , P_2 , P_3
The Gantt Chart for the schedule is:



- Waiting time for $P_1 = 0$; $P_2 = 24$; $P_3 = 27$
- Average waiting time: $(0 + 24 + 27)/3 = 17$





FCFS Scheduling (Cont.)

Suppose that the processes arrive in the order:

$$P_2, P_3, P_1$$

- The Gantt chart for the schedule is:



- Waiting time for $P_1 = 6$; $P_2 = 0$; $P_3 = 3$
- Average waiting time: $(6 + 0 + 3)/3 = 3$
- Much better than previous case
- **Convoy effect** - short process behind long process
 - Consider one CPU-bound and many I/O-bound processes





Shortest-Job-First (SJF) Scheduling

- Associate with each process the length of its next CPU burst
 - Use these lengths to schedule the process with the shortest time

- SJF is optimal – gives minimum average waiting time for a given set of processes
 - The difficulty is knowing the length of the next CPU request
 - Could ask the user





Example of SJF

Process

P_1

P_2

P_3

P_4

Burst Time

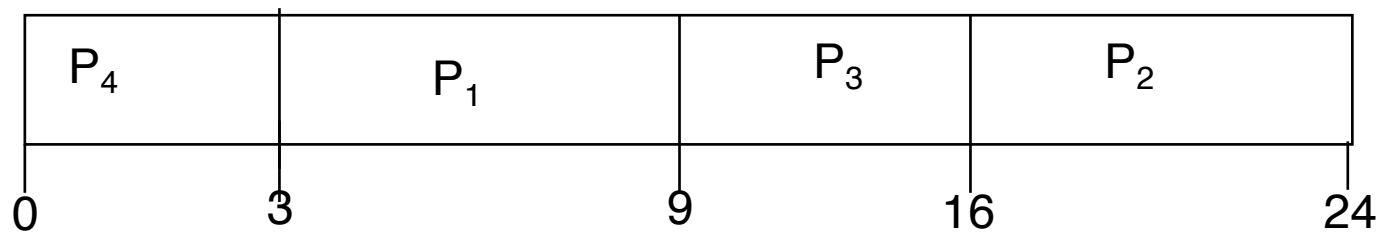
6

8

7

3

■ SJF scheduling chart



■ Average waiting time = $(3 + 16 + 9 + 0) / 4 = 7$

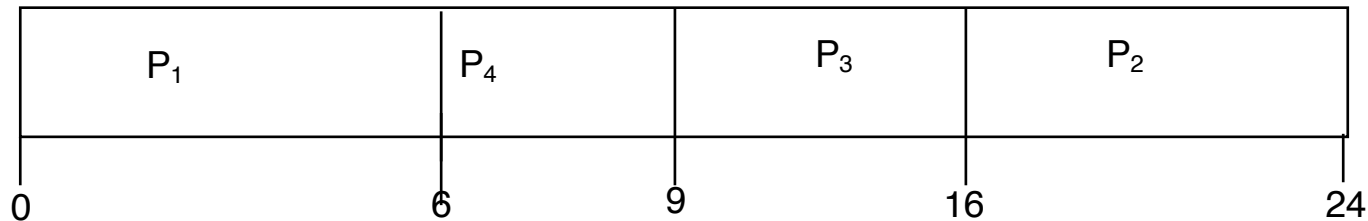




Example of Non-Preemptive SJF

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	6
P_2	1	8
P_3	2	7
P_4	3	3

■ SJF scheduling chart



■ Average waiting time = $(0 + 16 + 9 + 6) / 4 = 7.75$





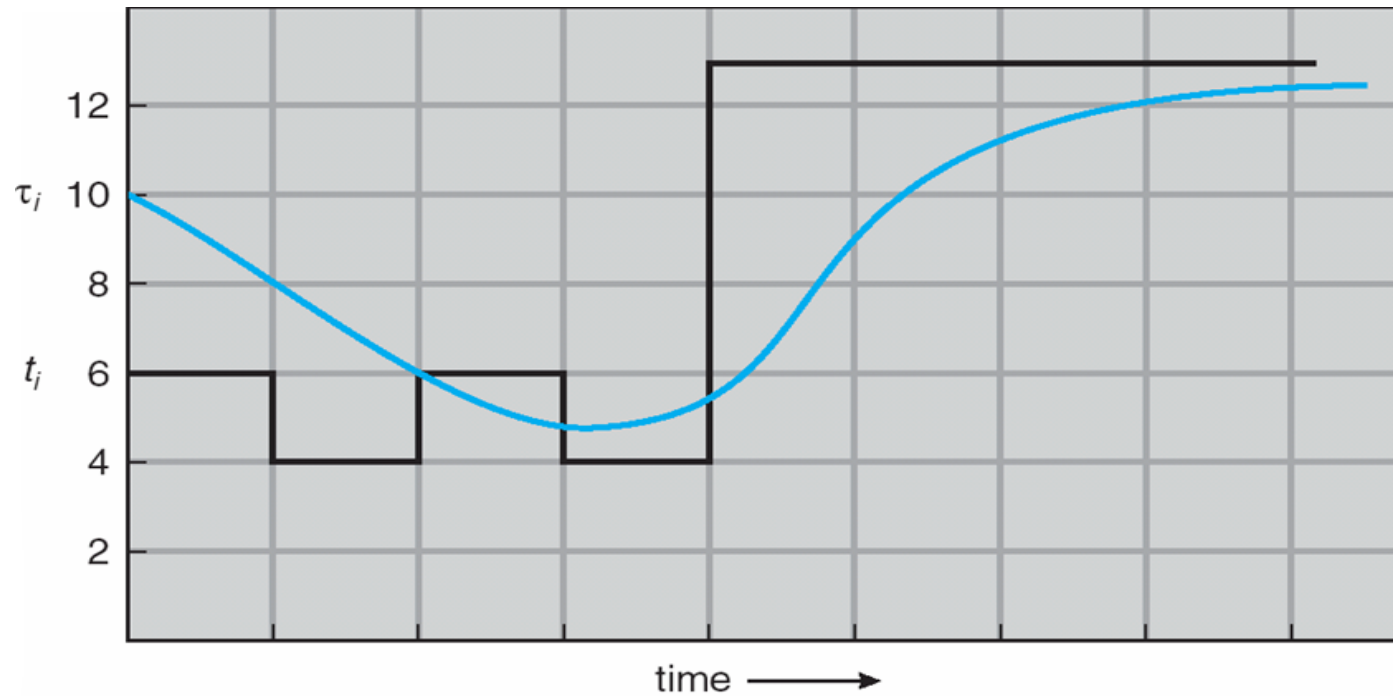
Determining Length of Next CPU Burst

- Can only estimate the length – should be similar to the previous one
 - Then pick process with shortest predicted next CPU burst
- Can be done by using the length of previous CPU bursts, using exponential averaging
 1. t_n = actual length of n^{th} CPU burst
 2. τ_{n+1} = predicted value for the next CPU burst
 3. $\alpha, 0 \leq \alpha \leq 1$
 4. Define :
- Commonly, α set to $\frac{1}{2}$ $\tau_{n+1} = \alpha t_n + (1 - \alpha)\tau_n$.
- Preemptive version called **shortest-remaining-time-first**





Prediction of the Length of the Next CPU Burst



CPU burst (t_i)	6	4	6	4	13	13	13	...	
"guess" (τ_i)	10	8	6	6	5	9	11	12	...





Examples of Exponential Averaging

- $\alpha = 0$

- $\tau_{n+1} = \tau_n$
- Recent history does not count

- $\alpha = 1$

- $\tau_{n+1} = \alpha t_n$
- Only the actual last CPU burst counts

- If we expand the formula, we get:

$$\begin{aligned}\tau_{n+1} = & \alpha t_n + (1 - \alpha)\alpha t_{n-1} + \dots \\ & + (1 - \alpha)^j \alpha t_{n-j} + \dots \\ & + (1 - \alpha)^{n+1} \tau_0\end{aligned}$$

- Since both α and $(1 - \alpha)$ are less than or equal to 1, each successive term has less weight than its predecessor



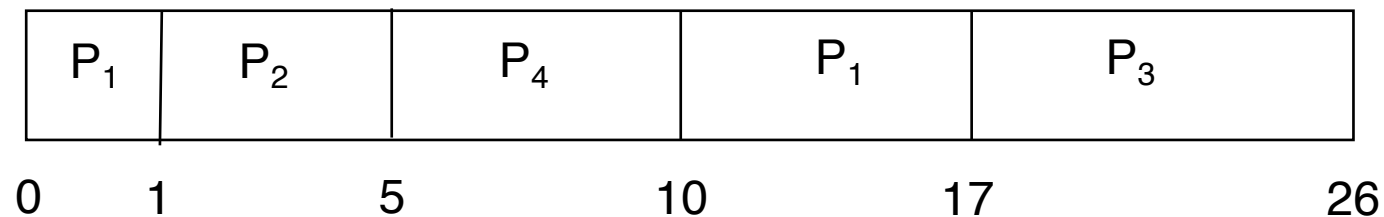


Example of Shortest-remaining-time-first

- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

- Preemptive* SJF Gantt Chart



- Average waiting time = $[(10-1)+(1-1)+(17-2)+5-3]/4 = 26/4 = 6.5$ msec





Exercise

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	1
P4	5	4

- Gantt for SRTF, FCFS
- Average waiting time?
- Average turnaround time?

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5





Priority Scheduling

- A priority number (integer) is associated with each process
- The CPU is allocated to the process with the highest priority (smallest integer $\equiv$ highest priority)
 - Preemptive
 - Nonpreemptive
- SJF is priority scheduling where priority is the inverse of predicted next CPU burst time
- Problem $\equiv$ **Starvation** – low priority processes may never execute
- Solution $\equiv$ **Aging** – as time progresses increase the priority of the process

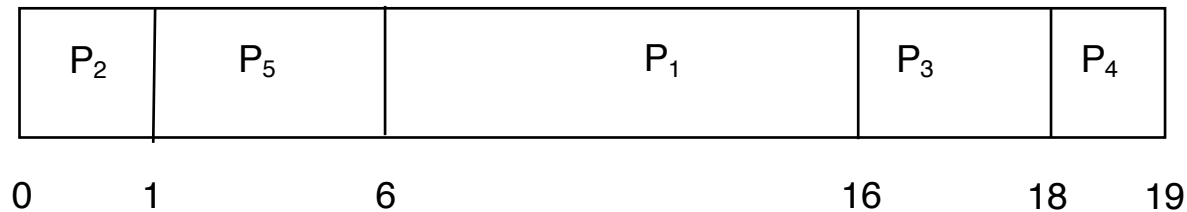




Example of Priority Scheduling

<u>Process</u>	<u>Burst Time</u>	<u>Priority</u>
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2

■ Priority scheduling Gantt Chart



■ Average waiting time = 8.2 msec





Round Robin (RR)

- Each process gets a small unit of CPU time (**time quantum** q), usually 10-100 milliseconds. After this time has elapsed, the process is preempted and added to the end of the ready queue.
- If there are n processes in the ready queue and the time quantum is q , then each process gets $1/n$ of the CPU time in chunks of at most q time units at once. No process waits more than $(n-1)q$ time units.
- Timer interrupts every quantum to schedule next process
- Performance
 - q large $\Rightarrow$ FIFO
 - q small $\Rightarrow q$ must be large with respect to context switch, otherwise overhead is too high

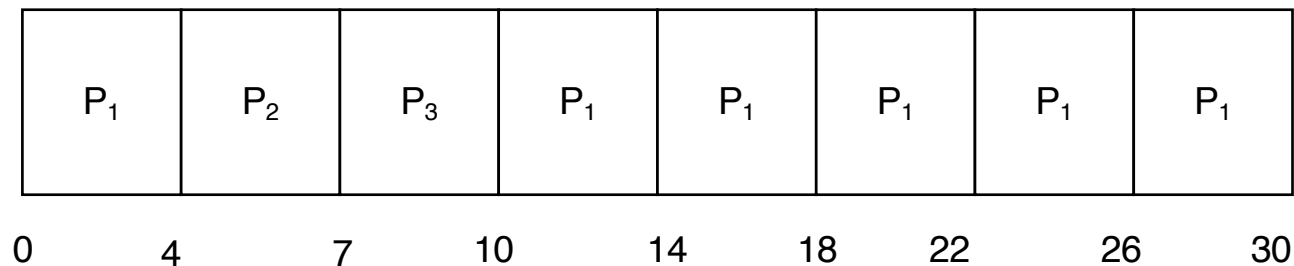




Example of RR with Time Quantum = 4

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- The Gantt chart is:



- Typically, higher average turnaround than SJF, but better *response*
- q should be large compared to context switch time
- q usually 10ms to 100ms, context switch < 10 usec





Exercise

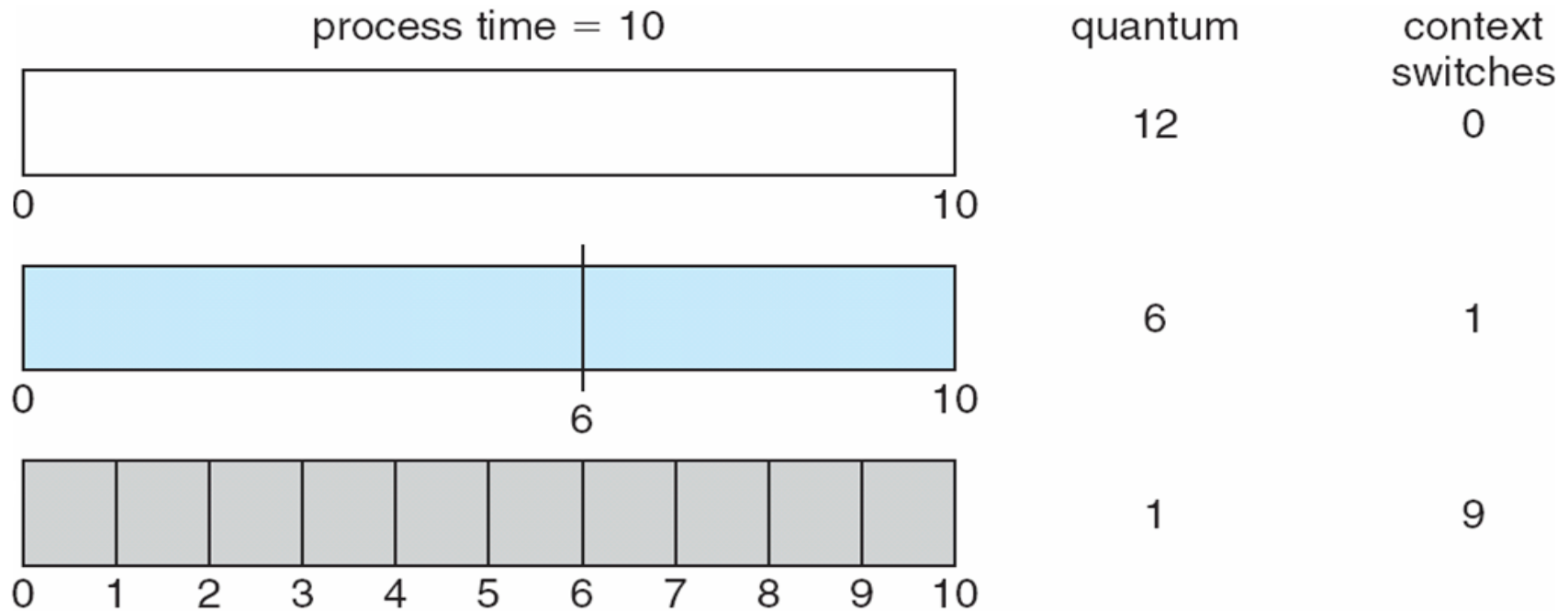
- Gantt for RR ($q=20$)
- Average waiting time?
- Average turnaround time?

Process	Burst Time
P1	53
P2	27
P3	68
P4	44
P5	38



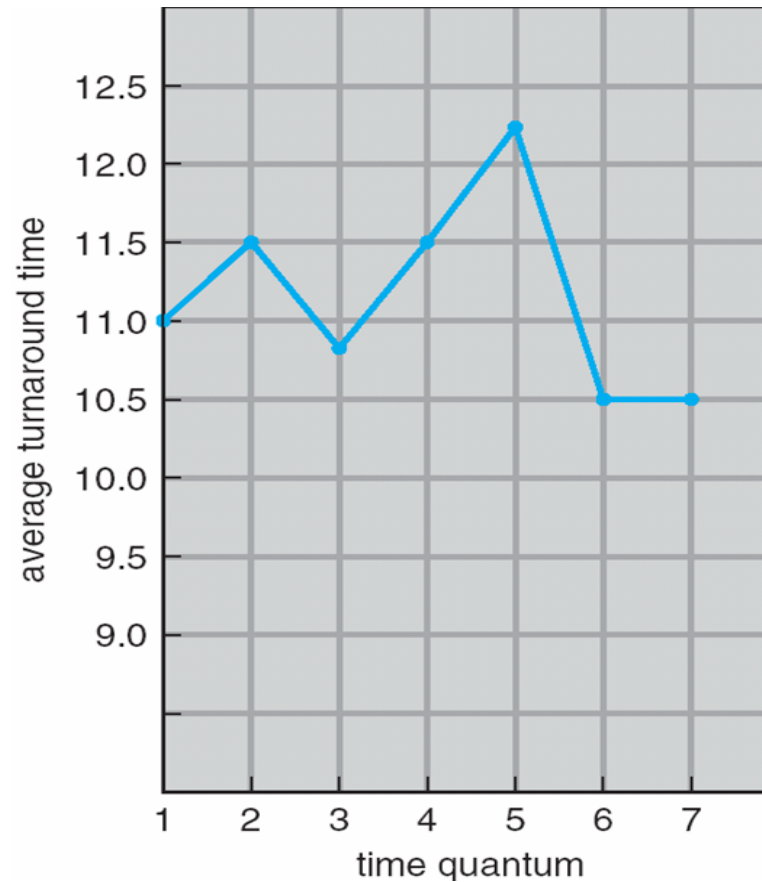


Time Quantum and Context Switch Time





Turnaround Time Varies With The Time Quantum



process	time
P_1	6
P_2	3
P_3	1
P_4	7

80% of CPU bursts should be shorter than q





Exercise

Process	Arrival Time	CPU Time
1	0	3
2	2	6
3	4	4
4	6	5
5	8	2

- Gantt for SJF Preemptive, RR ($q=3$)
- Average waiting time?
- Average turnaround time?





Multilevel Queue

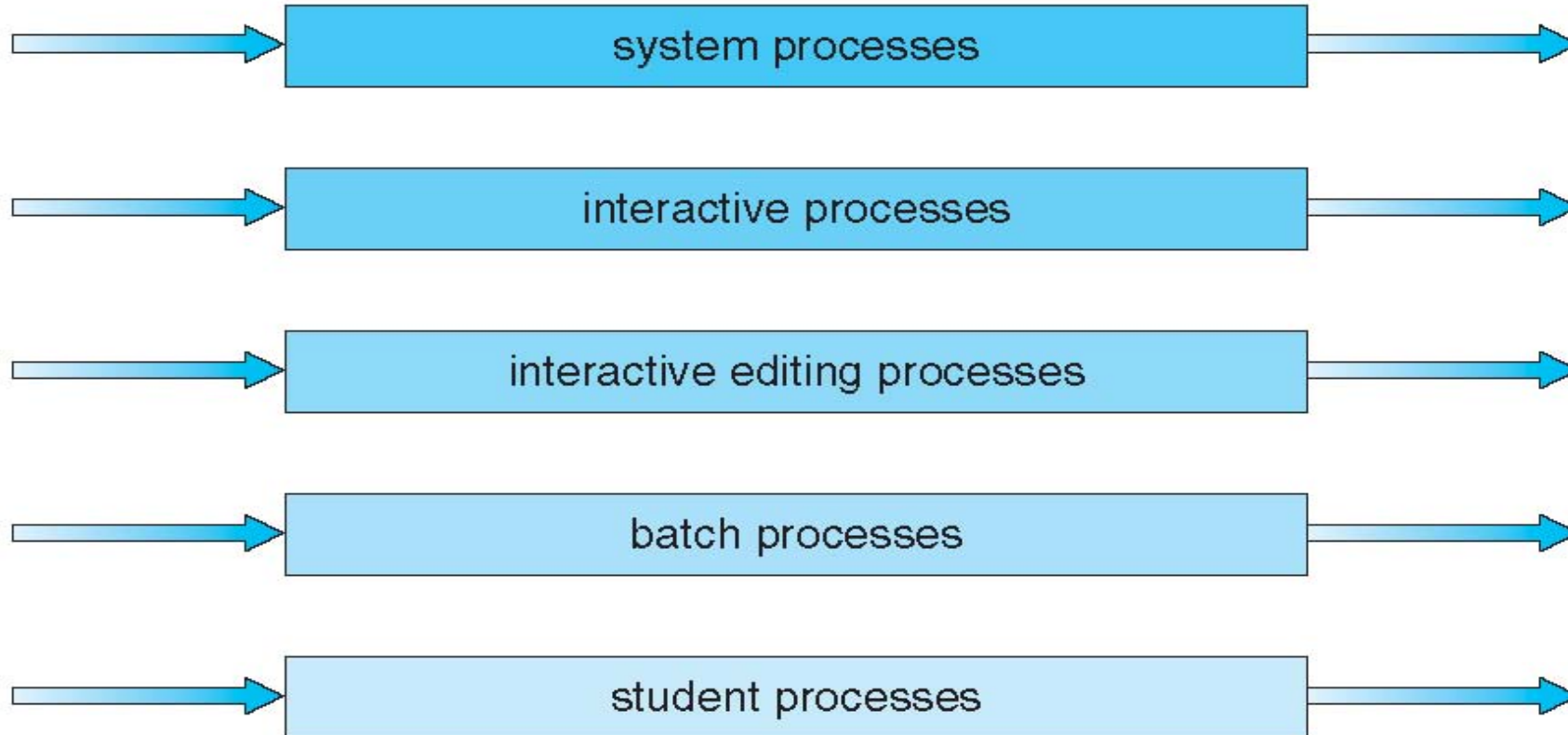
- Ready queue is partitioned into separate queues, eg:
 - foreground (interactive)
 - background (batch)
- Process permanently in a given queue
- Each queue has its own scheduling algorithm:
 - foreground – RR
 - background – FCFS
- Scheduling must be done between the queues:
 - Fixed priority scheduling; (i.e., serve all from foreground then from background). Possibility of starvation.
 - Time slice – each queue gets a certain amount of CPU time which it can schedule amongst its processes; i.e., 80% to foreground in RR
 - 20% to background in FCFS





Multilevel Queue Scheduling

highest priority



lowest priority





Multilevel Feedback Queue

- A process can move between the various queues; aging can be implemented this way
- Multilevel-feedback-queue scheduler defined by the following parameters:
 - number of queues
 - scheduling algorithms for each queue
 - method used to determine when to upgrade a process
 - method used to determine when to demote a process
 - method used to determine which queue a process will enter when that process needs service





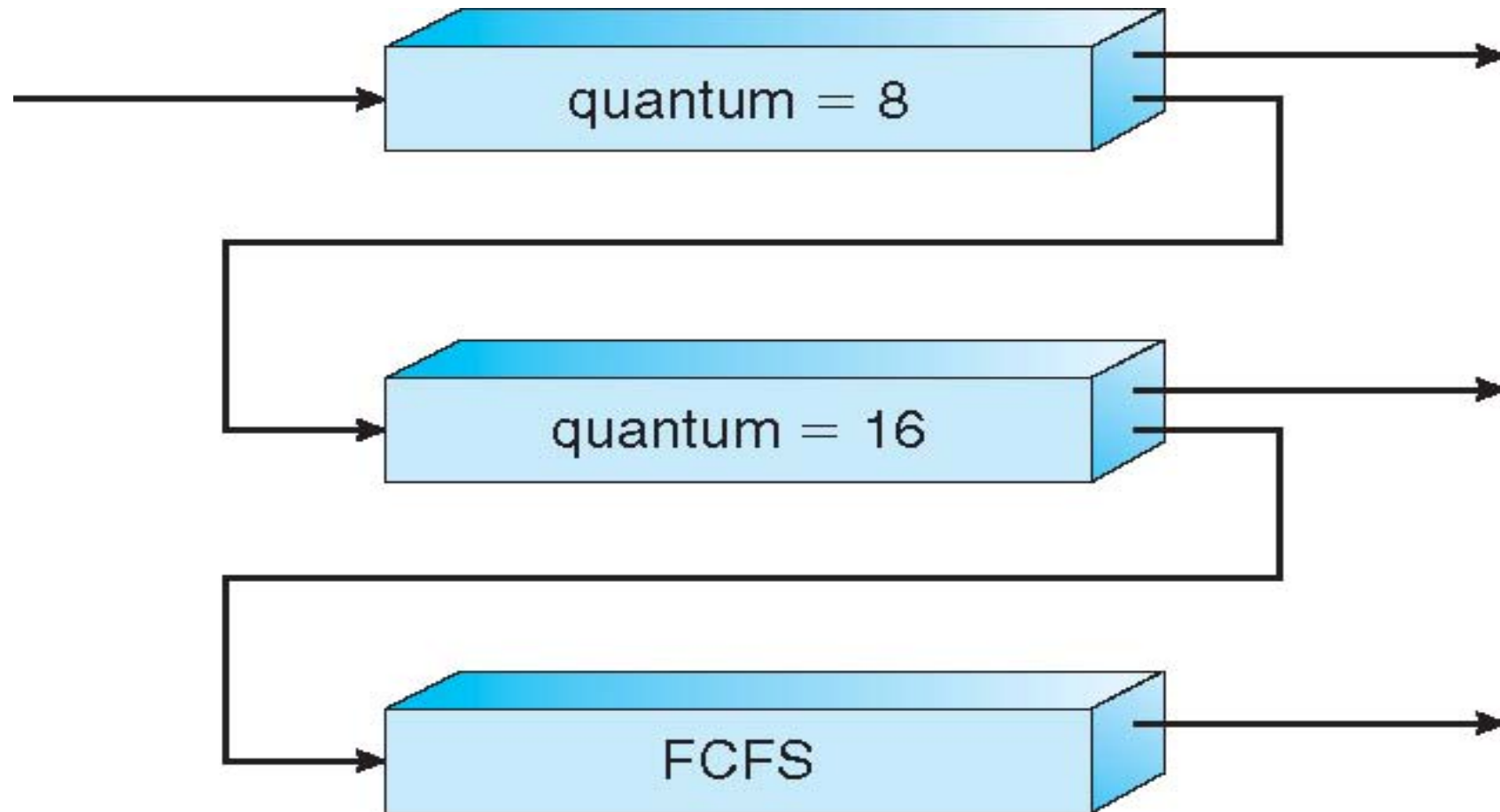
Example of Multilevel Feedback Queue

- Three queues:
 - Q_0 – RR with time quantum 8 milliseconds
 - Q_1 – RR time quantum 16 milliseconds
 - Q_2 – FCFS
- Scheduling
 - A new job enters queue Q_0 which is served FCFS
 - ▶ When it gains CPU, job receives 8 milliseconds
 - ▶ If it does not finish in 8 milliseconds, job is moved to queue Q_1
 - At Q_1 job is again served FCFS and receives 16 additional milliseconds
 - ▶ If it still does not complete, it is preempted and moved to queue Q_2





Multilevel Feedback Queues



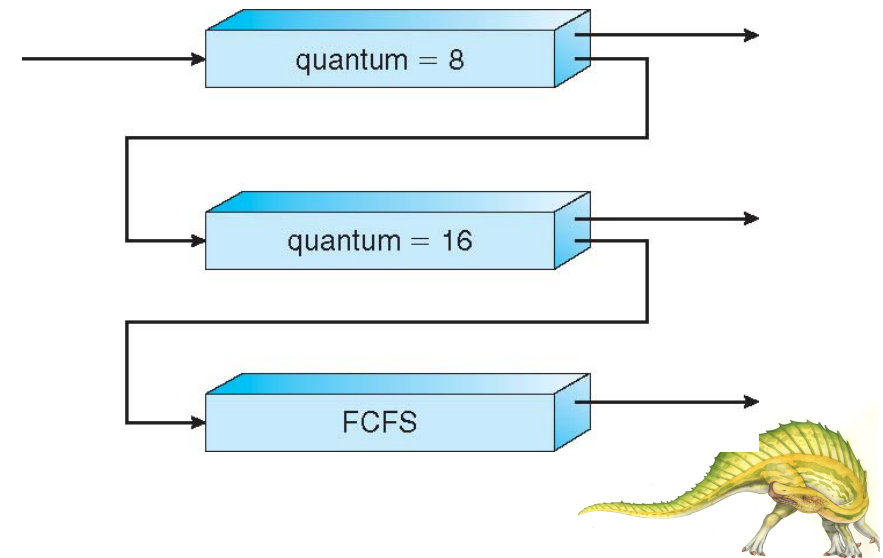


Example

Q_0 – RR with time quantum 8 milliseconds
 Q_1 – RR time quantum 16 milliseconds
 Q_2 – FCFS

Process	Burst Time
P1	53
P2	17
P3	68
P4	24

- ❑ Gantt for MFQ
- ❑ Average waiting time?
- ❑ Average turnaround time?





Exercise

Process	Burst - time	Priority
P1	10	3
P2	1	1 (highest)
P3	2	5
P4	1	4
p5	5	2

- Gantt for FCFS, SJF, RR ($q=1$), Priority
- Average waiting time?
- Average turnaround time?





Homework

- 6.3 p306
- 6.16 p308
- 6.17 p309





Thread Scheduling

- Distinction between user-level and kernel-level threads
- When threads supported, threads scheduled, not processes
- Many-to-one and many-to-many models, thread library schedules user-level threads to run on LWP
 - Known as **process-contention scope (PCS)** since scheduling competition is within the process
 - Typically done via priority set by programmer
- Kernel thread scheduled onto available CPU is **system-contention scope (SCS)** – competition among all threads in system





Pthread Scheduling

- API allows specifying either PCS or SCS during thread creation
 - PTHREAD_SCOPE_PROCESS schedules threads using PCS scheduling
 - PTHREAD_SCOPE_SYSTEM schedules threads using SCS scheduling
- Can be limited by OS – Linux and Mac OS X only allow PTHREAD_SCOPE_SYSTEM





Pthread Scheduling API

```
#include <pthread.h>
#include <stdio.h>
#define NUM_THREADS 5
int main(int argc, char *argv[])
{
    int i;
    pthread_t tid[NUM_THREADS];
    pthread_attr_t attr;
    /* get the default attributes */
    pthread_attr_t attr;
    pthread_attr_init(&attr);
    /* set the scheduling algorithm to PROCESS or SYSTEM */
    pthread_attr_setscope(&attr, PTHREAD_SCOPE_SYSTEM);
    /* set the scheduling policy - FIFO, RT, or OTHER */
    pthread_attr_setschedpolicy(&attr, SCHED_OTHER);
    /* create the threads */
    for (i = 0; i < NUM_THREADS; i++)
        pthread_create(&tid[i], &attr, runner, NULL);
}
```





Pthread Scheduling API

```
/* now join on each thread */
for (i = 0; i < NUM THREADS; i++)
    pthread join(tid[i], NULL);
}

/* Each thread will begin control in this
function */
void *runner(void *param)
{
    printf("I am a thread\n");
    pthread exit(0);
}
```





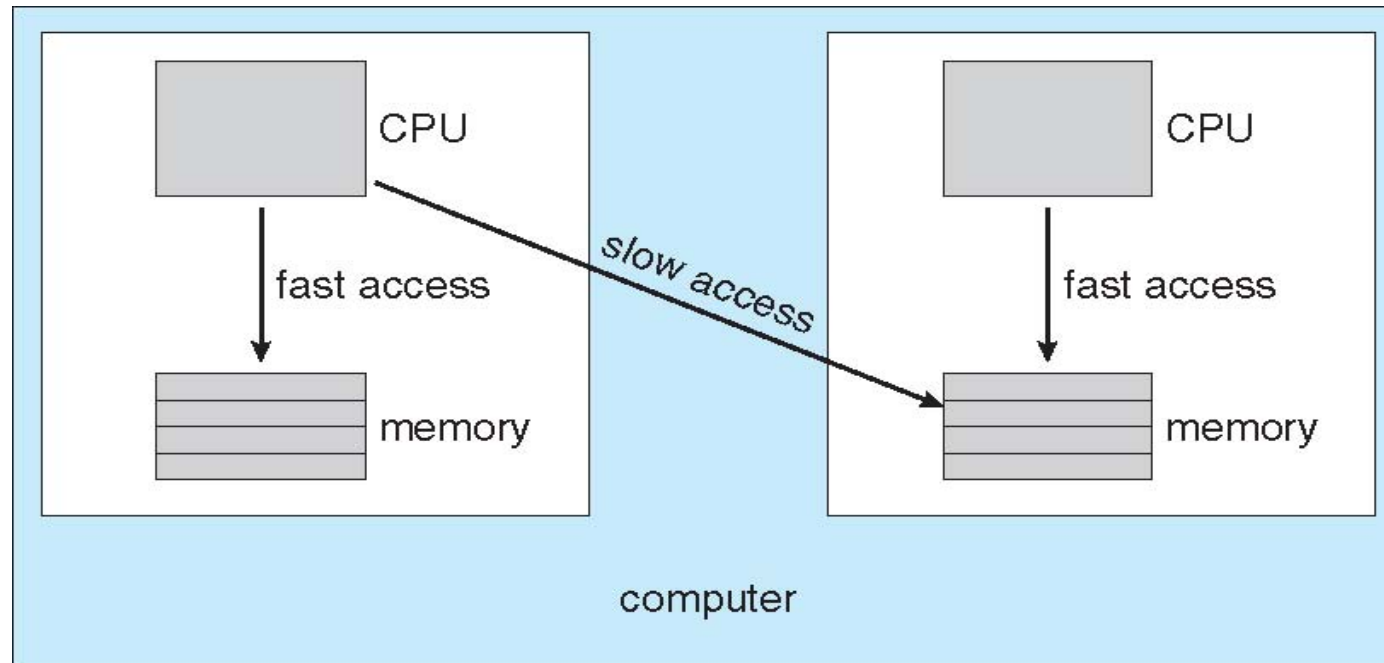
Multiple-Processor Scheduling

- CPU scheduling more complex when multiple CPUs are available
- **Homogeneous processors** within a multiprocessor
- **Asymmetric multiprocessing** – only one processor accesses the system data structures, alleviating the need for data sharing
- **Symmetric multiprocessing (SMP)** – each processor is self-scheduling, all processes in common ready queue, or each has its own private queue of ready processes
 - Currently, most common
- **Processor affinity** – process has affinity for processor on which it is currently running
 - **soft affinity**
 - **hard affinity**
 - Variations including **processor sets**





NUMA and CPU Scheduling



Note that memory-placement algorithms can also consider affinity





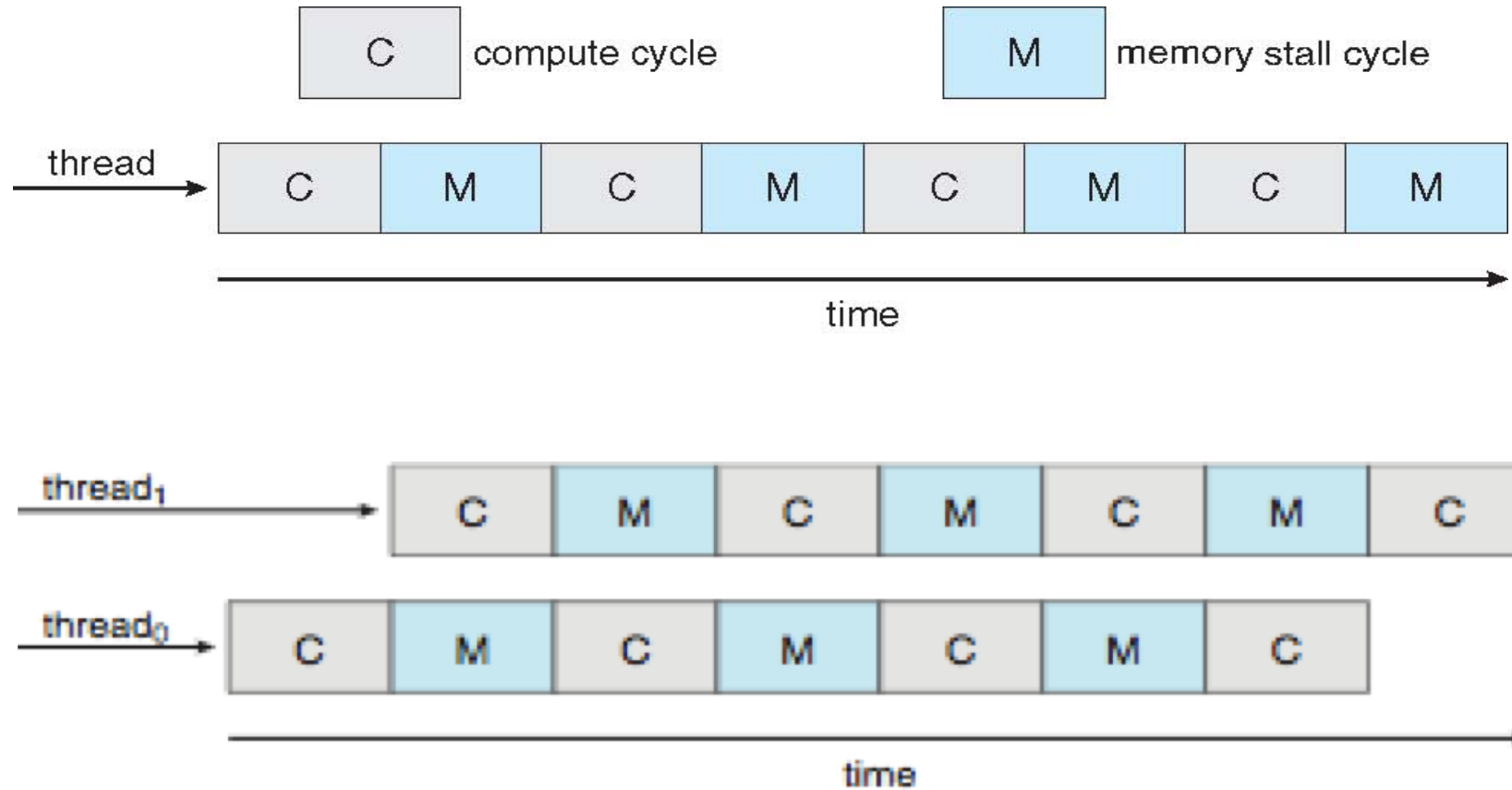
Multicore Processors

- Recent trend to place multiple processor cores on same physical chip
- Faster and consumes less power
- Multiple threads per core also growing
 - Takes advantage of memory stall to make progress on another thread while memory retrieve happens





Multithreaded Multicore System





Virtualization and Scheduling

- Virtualization software schedules multiple guests onto CPU(s)
- Each guest doing its own scheduling
 - Not knowing it doesn't own the CPUs
 - Can result in poor response time
 - Can effect time-of-day clocks in guests
- Can undo good scheduling algorithm efforts of guests





Operating System Examples

- Solaris scheduling
- Windows XP scheduling
- Linux scheduling





Solaris

- Priority-based scheduling
- Six classes available
 - Time sharing (default)
 - Interactive
 - Real time
 - System
 - Fair Share
 - Fixed priority
- Given thread can be in one class at a time
- Each class has its own scheduling algorithm
- Time sharing is multi-level feedback queue
 - Loadable table configurable by sysadmin





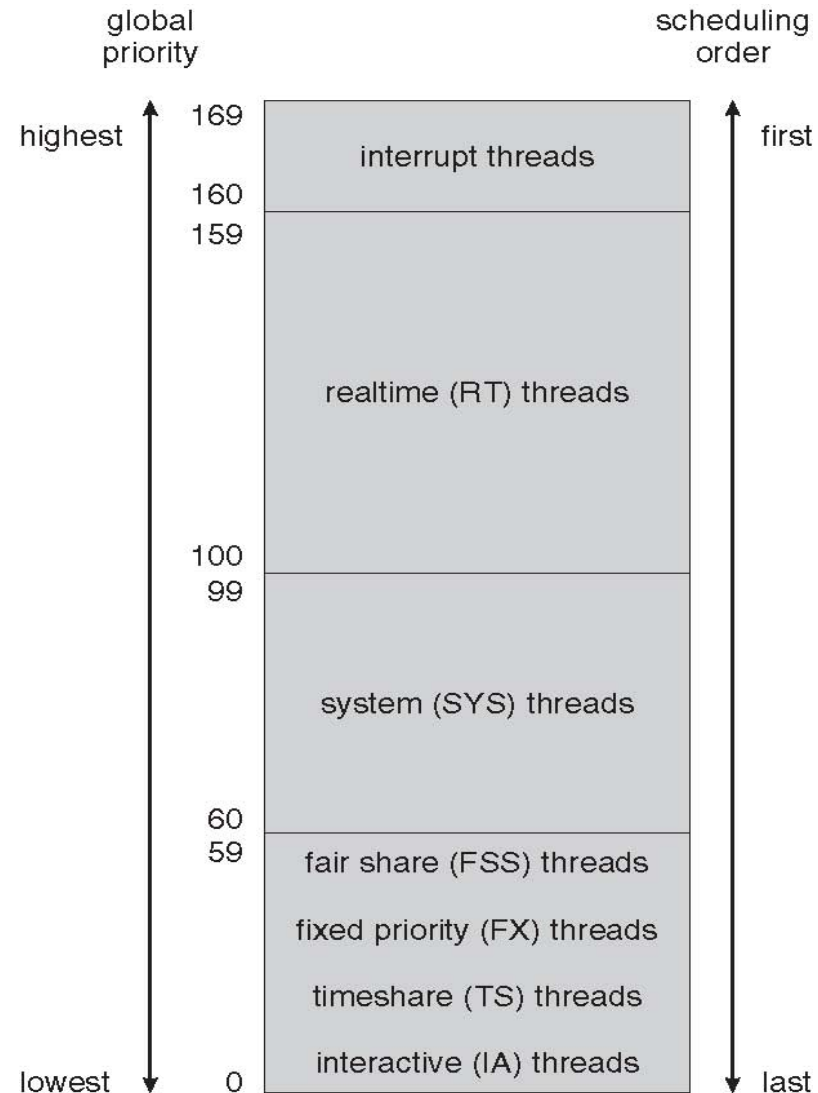
Solaris Dispatch Table

priority	time quantum	time quantum expired	return from sleep
0	200	0	50
5	200	0	50
10	160	0	51
15	160	5	51
20	120	10	52
25	120	15	52
30	80	20	53
35	80	25	54
40	40	30	55
45	40	35	56
50	40	40	58
55	40	45	58
59	20	49	59





Solaris Scheduling





Solaris Scheduling (Cont.)

- Scheduler converts class-specific priorities into a per-thread global priority
 - Thread with highest priority runs next
 - Runs until (1) blocks, (2) uses time slice, (3) preempted by higher-priority thread
 - Multiple threads at same priority selected via RR





Windows Scheduling

- Windows uses priority-based preemptive scheduling
- Highest-priority thread runs next
- *Dispatcher* is scheduler
- Thread runs until (1) blocks, (2) uses time slice, (3) preempted by higher-priority thread
- Real-time threads can preempt non-real-time
- 32-level priority scheme
- **Variable class** is 1-15, **real-time class** is 16-31
- Priority 0 is memory-management thread
- Queue for each priority
- If no run-able thread, runs **idle thread**





Windows Priority Classes

- Win32 API identifies several priority classes to which a process can belong
 - REALTIME_PRIORITY_CLASS, HIGH_PRIORITY_CLASS, ABOVE_NORMAL_PRIORITY_CLASS, NORMAL_PRIORITY_CLASS, BELOW_NORMAL_PRIORITY_CLASS, IDLE_PRIORITY_CLASS
 - All are variable except REALTIME
- A thread within a given priority class has a relative priority
 - TIME_CRITICAL, HIGHEST, ABOVE_NORMAL, NORMAL, BELOW_NORMAL, LOWEST, IDLE
- Priority class and relative priority combine to give numeric priority
- Base priority is NORMAL within the class
- If quantum expires, priority lowered, but never below base
- If wait occurs, priority boosted depending on what was waited for
- Foreground window given 3x priority boost





Windows XP Priorities

	real-time	high	above normal	normal	below normal	idle priority
time-critical	31	15	15	15	15	15
highest	26	15	12	10	8	6
above normal	25	14	11	9	7	5
normal	24	13	10	8	6	4
below normal	23	12	9	7	5	3
lowest	22	11	8	6	4	2
idle	16	1	1	1	1	1





Linux Scheduling

- Constant order $O(1)$ scheduling time
- Preemptive, priority based
- Two priority ranges: time-sharing and real-time
- **Real-time** range from 0 to 99 and **nice** value from 100 to 140
- Map into global priority with numerically lower values indicating higher priority
- Higher priority gets larger q
- Task run-able as long as time left in time slice (**active**)
- If no time left (**expired**), not run-able until all other tasks use their slices
- All run-able tasks tracked in per-CPU **runqueue** data structure
 - Two priority arrays (active, expired)
 - Tasks indexed by priority
 - When no more active, arrays are exchanged





Linux Scheduling (Cont.)

- Real-time scheduling according to POSIX.1b
 - Real-time tasks have static priorities
- All other tasks dynamic based on *nice* value plus or minus 5
 - Interactivity of task determines plus or minus
 - ▶ More interactive -> more minus
 - Priority recalculated when task expired
 - This exchanging arrays implements adjusted priorities





Priorities and Time-slice length

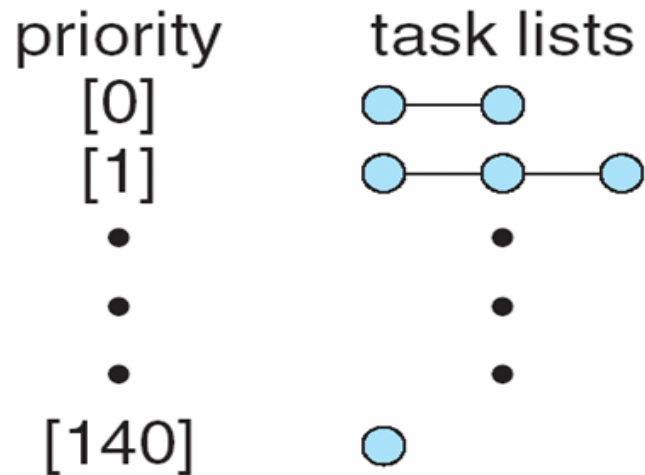
<u>numeric priority</u>	<u>relative priority</u>		<u>time quantum</u>
0	highest	real-time tasks	200 ms
•			
•			
•			
99			
100		other tasks	
•			
•			
•			
140	lowest		10 ms



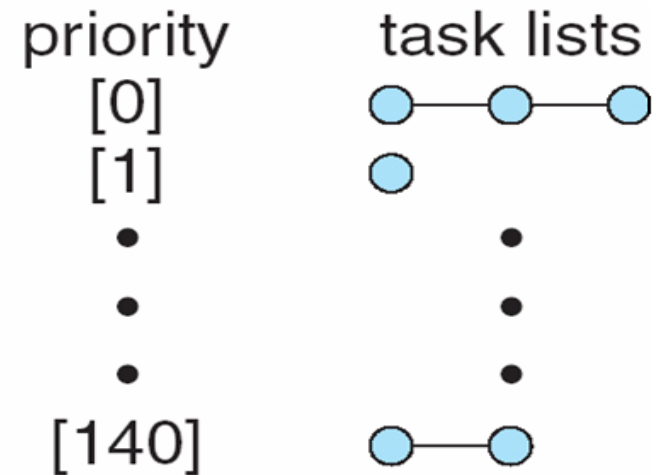


List of Tasks Indexed According to Priorities

active array



expired array





Algorithm Evaluation

- How to select CPU-scheduling algorithm for an OS?
- Determine criteria, then evaluate algorithms
- Deterministic modeling
 - Type of **analytic evaluation**
 - Takes a particular predetermined workload and defines the performance of each algorithm for that workload





Queueing Models

- Describes the arrival of processes, and CPU and I/O bursts probabilistically
 - Commonly exponential, and described by mean
 - Computes average throughput, utilization, waiting time, etc
- Computer system described as network of servers, each with queue of waiting processes
 - Knowing arrival rates and service rates
 - Computes utilization, average queue length, average wait time, etc





Little's Formula

- n = average queue length
- W = average waiting time in queue
- λ = average arrival rate into queue
- Little's law – in steady state, processes leaving queue must equal processes arriving, thus $n = \lambda \times W$
 - Valid for any scheduling algorithm and arrival distribution
- For example, if on average 7 processes arrive per second, and normally 14 processes in queue, then average wait time per process = 2 seconds





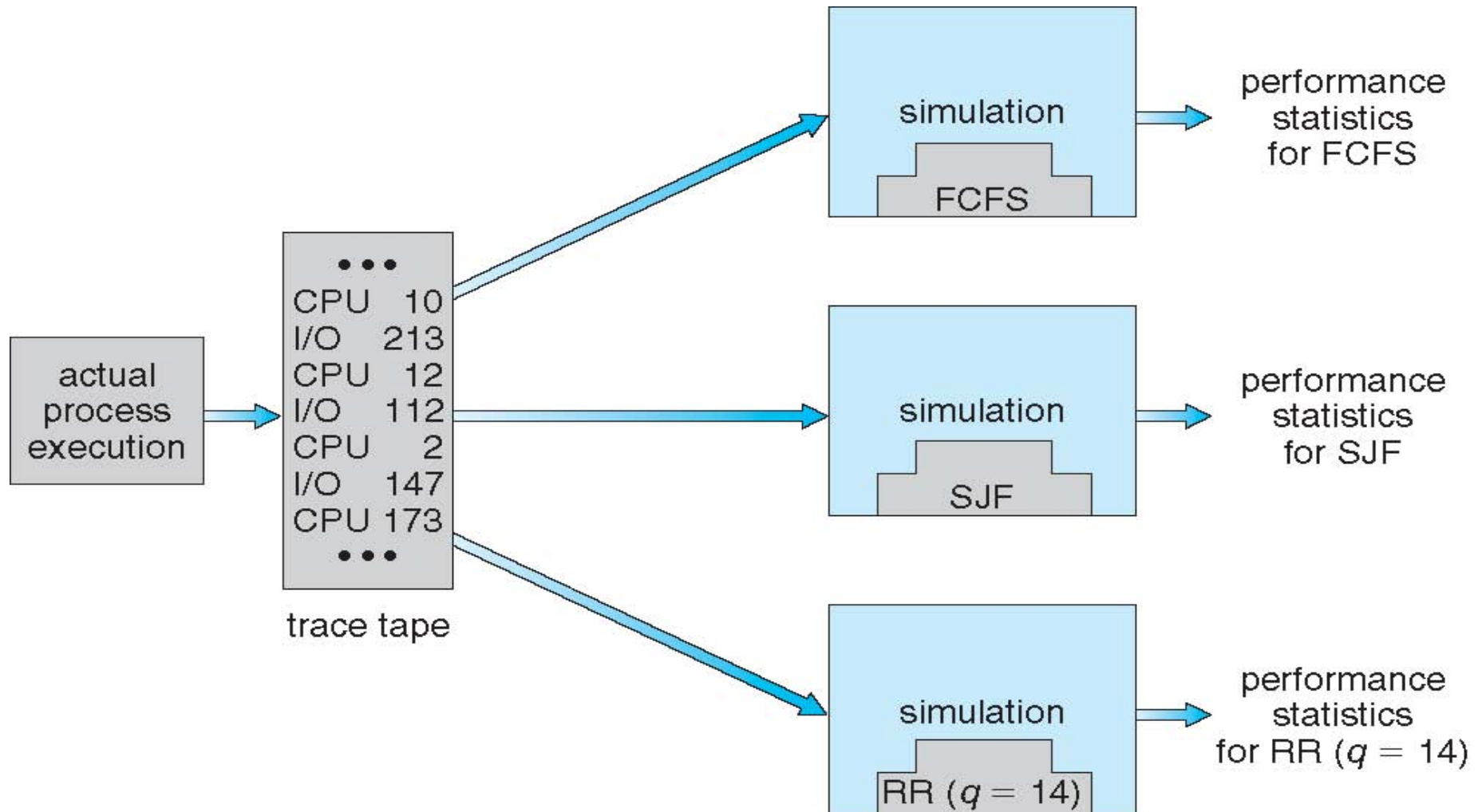
Simulations

- Queueing models limited
- **Simulations** more accurate
 - Programmed model of computer system
 - Clock is a variable
 - Gather statistics indicating algorithm performance
 - Data to drive simulation gathered via
 - ▶ Random number generator according to probabilities
 - ▶ Distributions defined mathematically or empirically
 - ▶ Trace tapes record sequences of real events in real systems





Evaluation of CPU Schedulers by Simulation



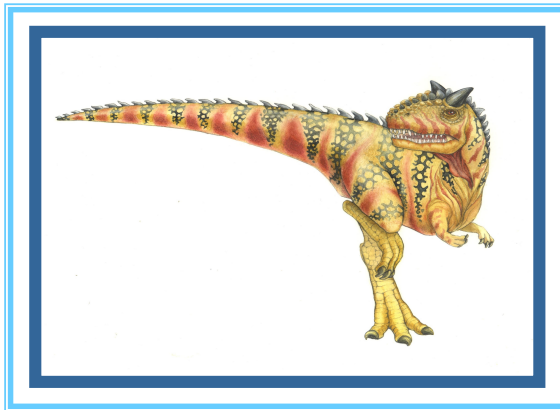


Implementation

- Even simulations have limited accuracy
- Just implement new scheduler and test in real systems
 - High cost, high risk
 - Environments vary
- Most flexible schedulers can be modified per-site or per-system
- Or APIs to modify priorities
- But again environments vary

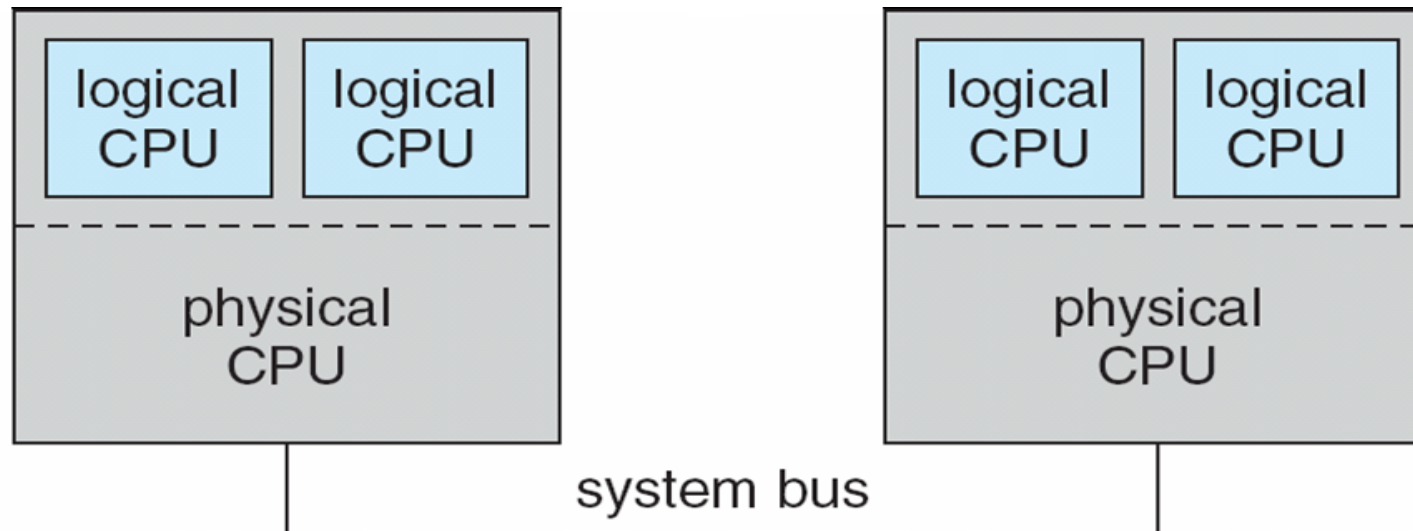


End of Chapter 5



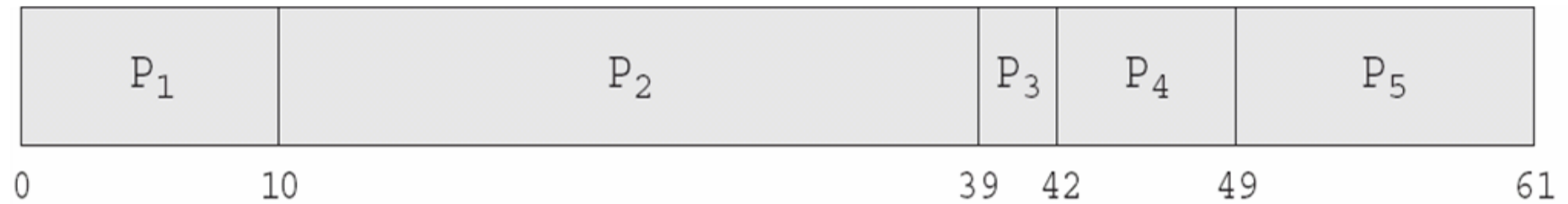


5.08



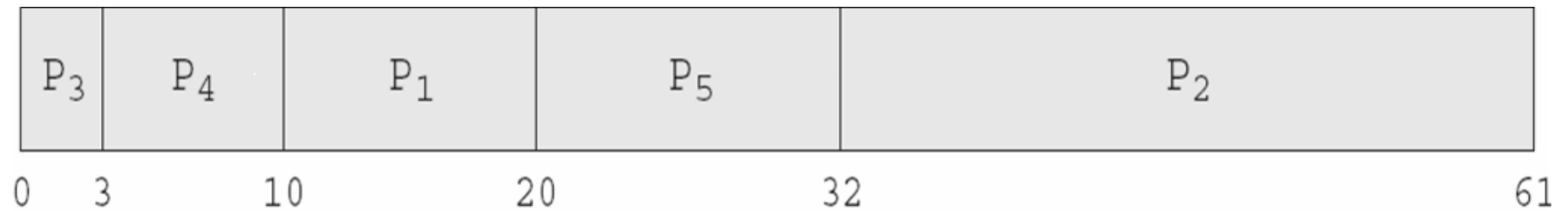


In-5.7



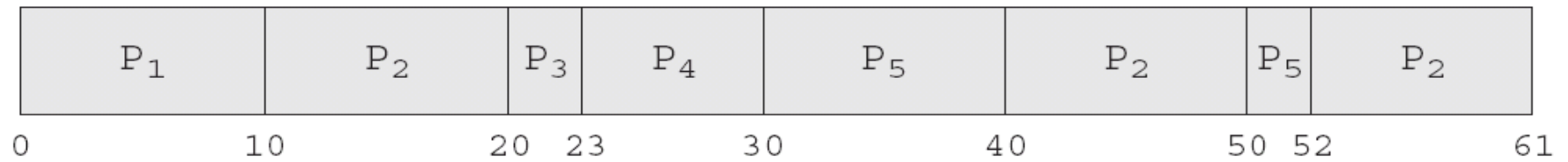


In-5.8



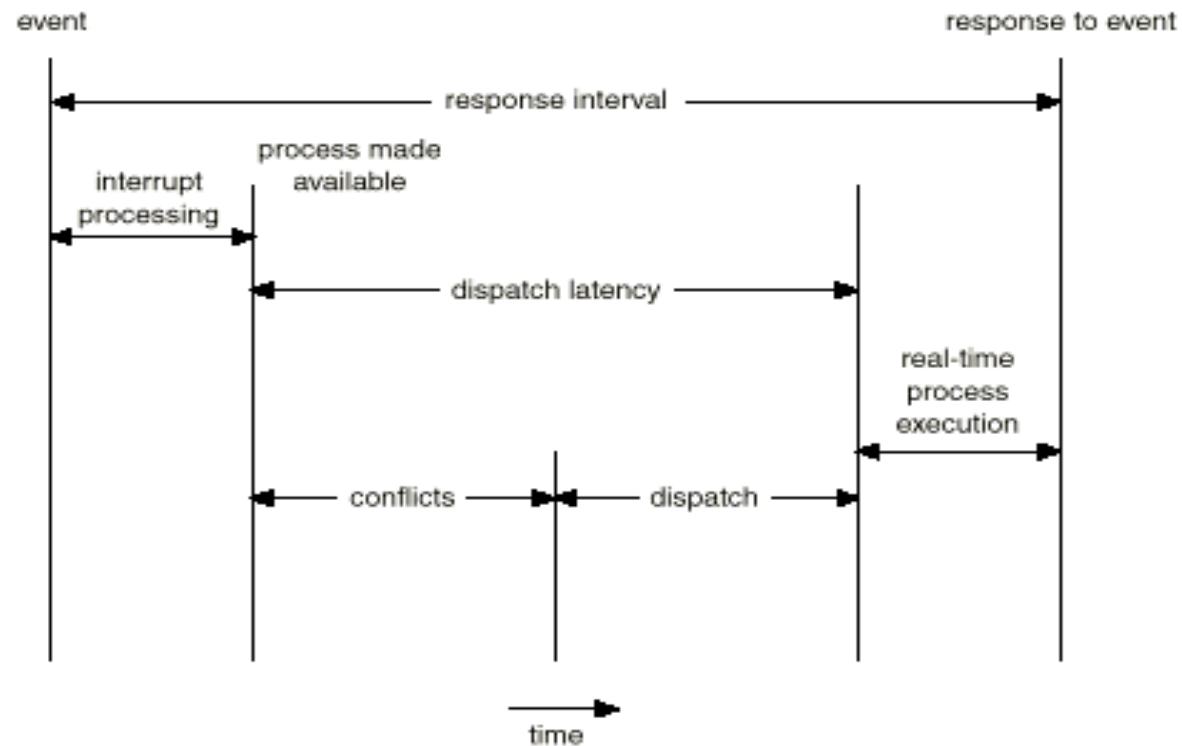


In-5.9





Dispatch Latency





Java Thread Scheduling

- JVM Uses a Preemptive, Priority-Based Scheduling Algorithm
- FIFO Queue is Used if There Are Multiple Threads With the Same Priority





Java Thread Scheduling (Cont.)

JVM Schedules a Thread to Run When:

1. The Currently Running Thread Exits the Runnable State
2. A Higher Priority Thread Enters the Runnable State

* Note – the JVM Does Not Specify Whether Threads are Time-Sliced or Not





Time-Slicing

Since the JVM Doesn't Ensure Time-Slicing, the `yield()` Method May Be Used:

```
while (true) {  
    // perform CPU-intensive task  
    . . .  
    Thread.yield();  
}
```

This Yields Control to Another Thread of Equal Priority





Thread Priorities

<u>Priority</u>	<u>Comment</u>
Thread.MIN_PRIORITY	Minimum Thread Priority
Thread.MAX_PRIORITY	Maximum Thread Priority
Thread.NORM_PRIORITY	Default Thread Priority

Priorities May Be Set Using `setPriority()` method:

```
setPriority(Thread.NORM_PRIORITY + 2);
```





Solaris 2 Scheduling

